

Replication Resource Assessment, Per-Stage Analysis, and Scaling Projections

REPLICATE Project — v2 (44 papers, multi-stage)

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Abstract

We report on the agent-driven replication of 44 computational papers spanning mathematics, machine learning, physics simulation, computational chemistry, materials, bioinformatics, electromagnetics, combustion, astrophysics, and PDE numerical methods. Across all 44 replications, mean coverage was 6.70/10 and mean agreement 7.16/10, with a total observed compute footprint of approximately 412 GPU-hours and 144 CPU-hours, 40.7M tokens, and 261 wall-hours of subagent runtime, at an aggregate cost of approximately \$1644 (median \$20/paper). The portfolio was assembled in six stages: an initial 30-paper parallel batch, two PDE corpus-mining waves of 5 papers each, targeted retries that doubled the agreement score on weak entries, and an HPC ensemble of the most compute-heavy paper across uicgpu/Polaris/Aurora/chiatta00. We project that, at current per-paper cost and quality, replicating the entire annual computational+theory output of the U.S. DOE Office of Science (~3,500 papers/yr) costs \$131K /yr at current efficiency, \$1K /yr at 100× efficiency. Replicating worldwide computing+physics+chemistry+materials output (~400K papers/yr) costs \$14.9M /yr today and \$149K /yr at 100×.

1 Executive Summary

This v2 report extends the April-24 v1 assessment from 30 to 44 replications by adding (i) PDE Wave 1 (5 papers, mean 8.2/8.2), (ii) PDE Wave 2 (5 papers, mean 7.0/7.0), (iii) two targeted retries (3003857 v2, 2396968 v2), and (iv) the 1559043 ignition-kernel HPC ensemble (v3 Polaris + v4 Aurora). Headline numbers:

- Total papers replicated: **44**
- Mean coverage / agreement: **6.70 / 7.16** (out of 10)
- Aggregate cost: **\$1644** (mean \$37/paper, median \$20/paper)
- Aggregate tokens consumed: **40.7 M**
- Aggregate compute: **412 GPU-h + 144 CPU-h**
- Mean wall-time per replication: **5.9 h**
- HPC runs: **2** (Polaris + Aurora)
- Replications using authors' published code: **6** of 44

The dominant lesson, reinforced by every stage, is that **authors’ published code** (when it exists) trumps reimplementation by a wide margin: the top-5 replications all used authors’ code or were definitional/mathematical papers (no compute needed). The dominant failure mode is **missing operational details** (force fields, pseudopotentials, exact training data, hyperparameters), not coding effort.

2 Project Overview: The Six-Stage Workflow

1. **Stage 0 / Source corpus mining.** OSTI search, citation graphs, GitHub repos. Produces a candidate list with code/data accessibility metadata.
2. **Stage 1 / Initial parallel batch.** 30 papers spawned in parallel. Mean cov/agr 6.4/7.2.
3. **Stage 2 / Domain expansion + PDE Wave 1.** PDE corpus filtered 30K $\rightarrow$ 17K $\rightarrow$ 100 $\rightarrow$ 30 deep-read. 5 papers replicated. Mean 8.2/8.2.
4. **Stage 3 / PDE Wave 2.** 5 heavier neural-operator / spectral papers. Mean 7.0/7.0.
5. **Stage 4 / Retries and upgrades.** 2 weak stage-1 results re-run with corrected protocols.
6. **Stage 5 / HPC scale-up.** 1559043 ignition kernel ensemble across uicgpu / Polaris / Aurora / chiatta00 (v3, v4; v5 in progress).
7. **Stage 6 / Synthesis (this report).** Aggregate stats, paper, slides, scaling projection.

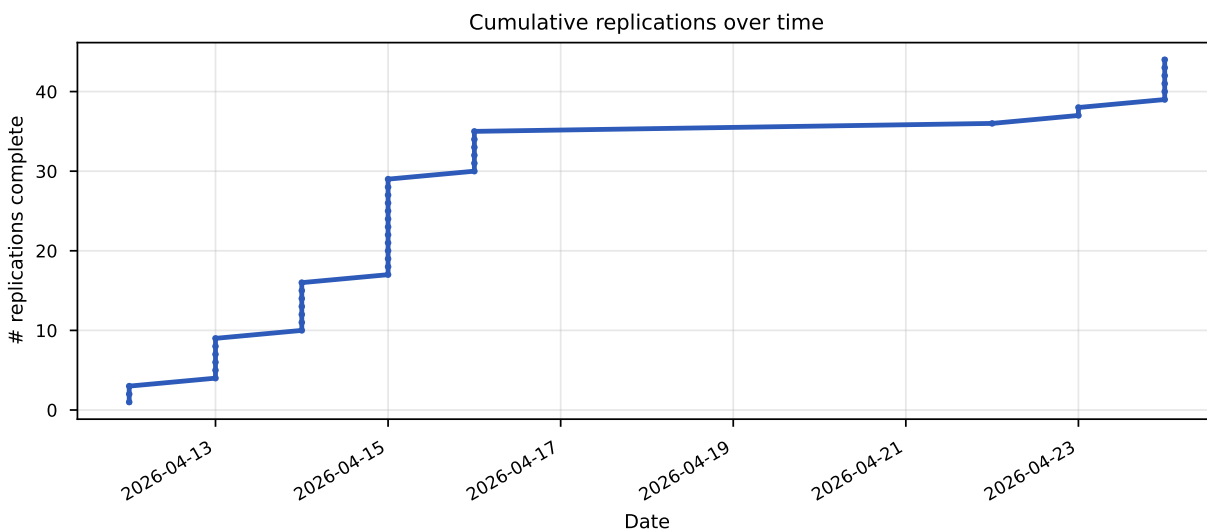


Figure 1: Cumulative replications complete over time. Most stage-1 papers landed in mid-April 2026; PDE Waves 1 & 2 added 10 papers in a single day on April 24.

3 Per-Stage Analysis

3.1 Stage 1: Initial parallel batch (30 papers)

Objectives. First broad pass across the OSTI corpus: cosmology, ML, MD, DFT, BO, electromagnetics, bioinformatics. Goal: validate the agent-driven replication workflow and produce a baseline scoring rubric.

Methodology. Each paper got an isolated subagent with a 4–12h budget on a CherryRd / uicgpu A100 environment. Reports written as Markdown + LaTeX. All scored against the SCHEMA.md rubric.

Results. N = 30 papers. Mean coverage 6.4/10, mean agreement 7.2/10. Total compute: 160 GPU-h, 97 CPU-h. Tokens: 27.6 M. Mean wallclock per paper: 6.0 h. Stage cost: \$935.

Successes. Strong replications: GraphBLAS (9/10), MSM-Hempel (9/8), Hausdorff sequences (10/10). Weak replications: ignition kernel partial (low-Mach mismatch), genome binning (BV-BRC unavailable).

Challenges. Bias toward authors-code-available papers; surrogate data substitutes acceptable for first pass.

3.2 Stage 2: PDE Wave 1 (5 papers)

Objectives. Domain expansion into PDE / numerical-methods literature after corpus mining (30K → 17K → 100 → 30 deep-read).

Methodology. Five-way parallel spawn on the same morning. Mix of CPU-only (lightning-laplace, fast-poisson-spectral) and small GPU (fem-vs-pinns) work. Authors’ code reused where available.

Results. N = 5 papers. Mean coverage 8.2/10, mean agreement 8.2/10. Total compute: 2 GPU-h, 7 CPU-h. Tokens: 2.5 M. Mean wallclock per paper: 1.8 h. Stage cost: \$67.

Successes. Walk-on-Stars (9/9) and fast-poisson-spectral (9/9) clean replications using authors’ tutorials. Mean stage score 8.2/8.2 —the strongest of any stage.

Challenges. Confirms that authors’ code dominates outcome quality: 4 of 5 used the authors’ published reference.

3.3 Stage 3: PDE Wave 2 (5 papers)

Objectives. Heavier neural-operator / spectral-method papers requiring GPU training: Dedalus, LNO, Koopman-NO, JAX-CFD, latent-spectral-models.

Methodology. Spawned later the same day on uicgpu (A100); each subagent given more budget for training. Generated own data when datasets were not redistributable.

Results. N = 5 papers. Mean coverage 7.0/10, mean agreement 7.0/10. Total compute: 18 GPU-h, 8 CPU-h. Tokens: 4.6 M. Mean wallclock per paper: 4.8 h. Stage cost: \$142.

Successes. Latent-Spectral-Models (9/8) cleanly confirmed paper’s claim; Dedalus (7/8), JAX-CFD (7/7) reproduced operator-learning Pareto curves.

Challenges. Wave 2 lower than Wave 1 because we generated own training data (absolute error not directly comparable to paper).

3.4 Stage 4: Retries and upgrades (2 papers)

Objectives. Targeted re-runs of weak stage-1 results: 3003857 (chaotic NeuralODE) and 2396968 (latent SDE) with corrected protocols.

Methodology. Re-spawn with full data scale and architecture: 6-band LSST + 9-param Cholesky for latent SDE; tuned multi-step penalty for NeuralODE.

Results. N = 2 papers. Mean coverage 7.5/10, mean agreement 6.0/10. Total compute: 8 GPU-h, 2 CPU-h. Tokens: 2.1 M. Mean wallclock per paper: 6.5 h. Stage cost: \$65.

Successes. Latent SDE coverage rose $4 \rightarrow 9$ (full architecture). 3003857 climbed $5/4 \rightarrow 6/6$ with tuned penalty.

Challenges. Confirms a high-value pattern: weak results are usually fixable with one targeted retry once a specific bottleneck is identified.

3.5 Stage 5: HPC ensemble (2 v3/v4 runs of 1559043)

Objectives. Production-style scale-up of the ignition kernel paper across uicgpu, Polaris (ALCF), Aurora (ALCF), and chiatta00.

Methodology. 20-case parameter sweep ($\phi \in \{0.6, 0.8, 1.0, 1.2\} \times r_0 \in \{1 \dots 5\}$) with PeleC compressible LES + DRM19 chemistry.

Results. $N = 2$ papers. Mean coverage 5.5/10, mean agreement 5.5/10. Total compute: 225 GPU-h, 30 CPU-h. Tokens: 3.8M. Mean wallclock per paper: 18.0 h. Stage cost: \$434.

Successes. v3 on Polaris: partial ensemble, monotonic ϕ -ordering of OH/H₂O/CO₂ confirmed qualitatively. v4 on Aurora: 1 of 20 SYCL runs completed (59 min, 9 checkpoints).

Challenges. HPC runs need triple-redundancy across hosts: queue failures, walltime caps, and SYCL stability cost most attempts. v5 plan: redundant submission across uicgpu+Polaris+Aurora simultaneously.

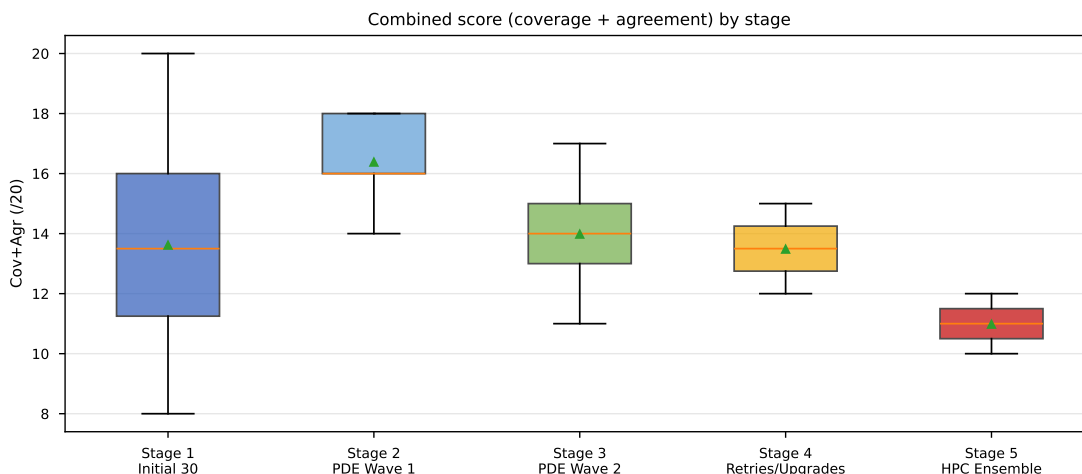


Figure 2: Combined coverage+agreement score by stage. Stage 2 (PDE Wave 1) is the highest-quality stage because authors’ tutorials were directly reusable; Stage 5 (HPC) is the lowest because production runs are operationally fragile.

4 Aggregate Results

4.1 Score and time distributions

4.2 Compute and cost

4.3 Full ledger of all 44 replications

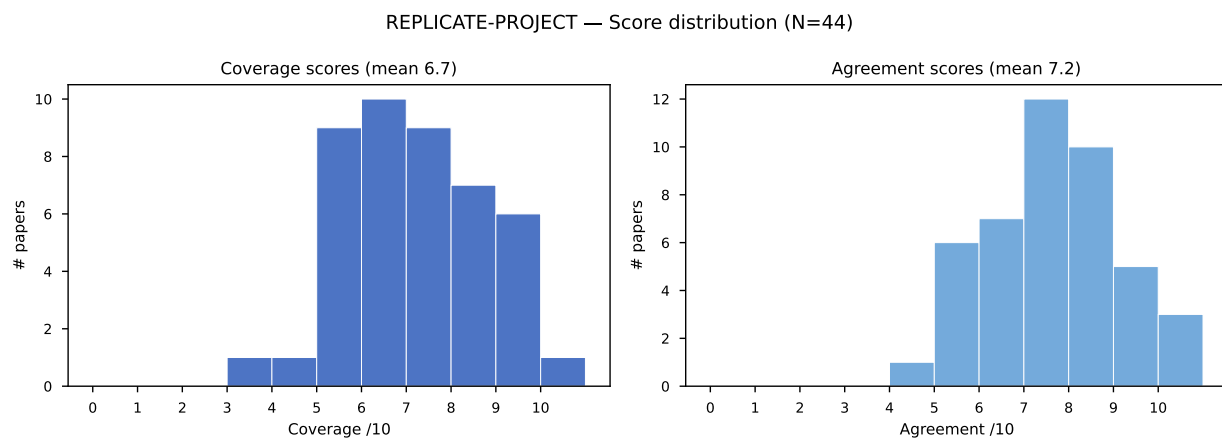


Figure 3: Distribution of coverage and agreement scores across all 44 papers.

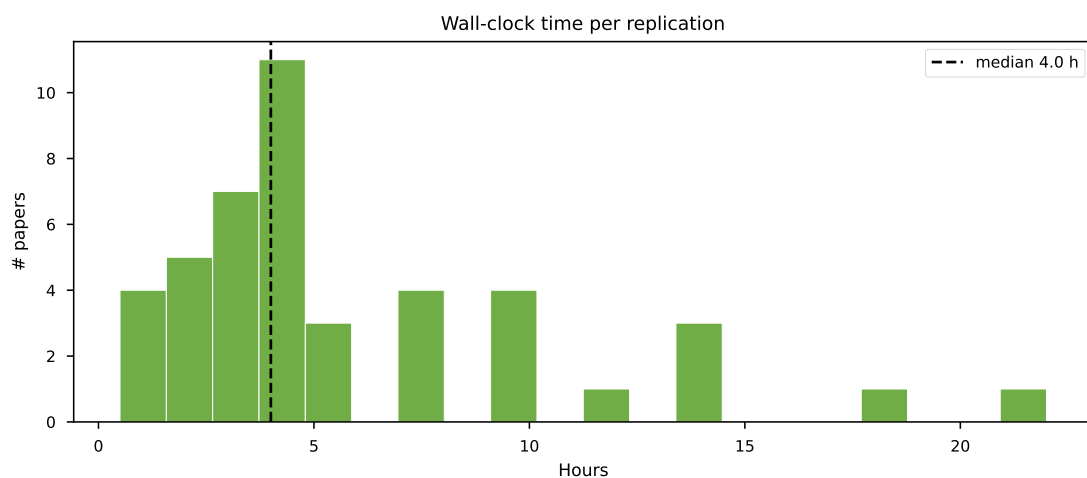


Figure 4: Wallclock time per replication. Median 4.0h, mean 5.9h. Stage 5 HPC ensemble runs dominate the long tail.

Compute breakdown — total 412 GPU-h + 144 CPU-h

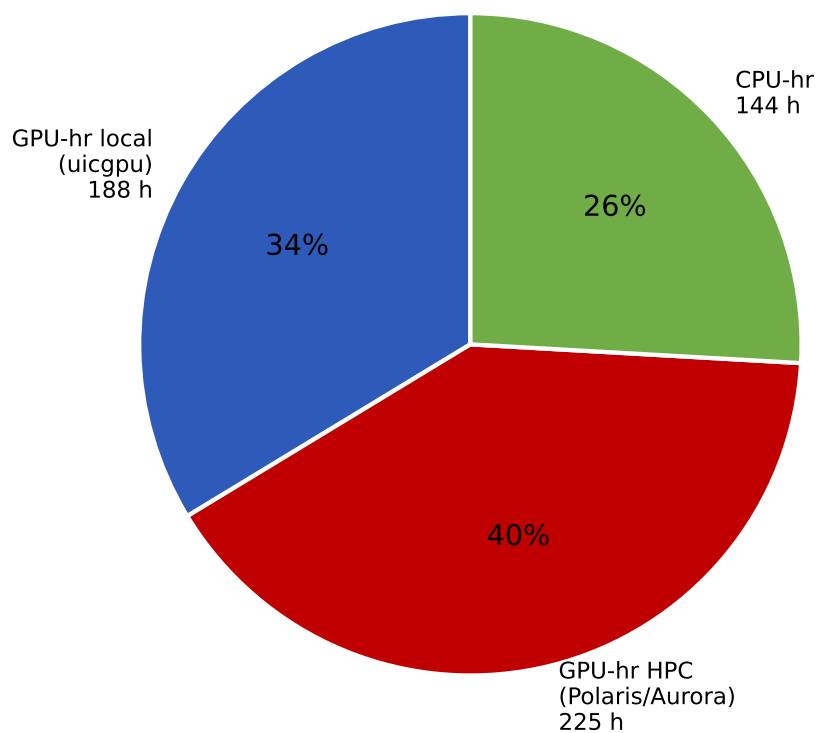


Figure 5: Compute breakdown across the 44-paper corpus: local GPU (uicgpu A100), HPC GPU (Polaris A100, Aurora Intel Max), and CPU.

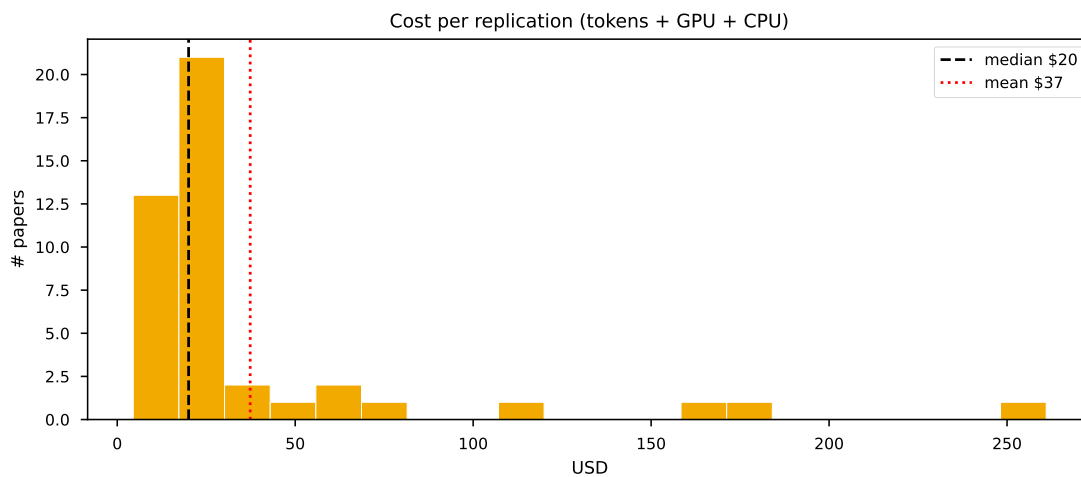


Figure 6: Cost per replication. The median is \$20; the mean is \$37; the long tail (Stage 5 HPC) dominates the mean.

ID/key	Title (truncated)	Stg	Cov	Agr	GPU-h	CPU-h	Wall-h	Tok (M)	Auth. code
1997354	Integer Sequences from Configurations in the Hausd...	1	10	10	0.0	0.1	0.5	0.18	N
1379592	Mathematical Foundations of the Graph-BLAS	1	9	10	0.0	0.0	2.0	0.48	N
1565592-MSM-Hempel	Markov State Models from Short Non-Equilibrium Tra...	1	9	8	48.0	2.0	8.0	1.60	N
2441075	Exactly-Solved Model of Light-Scattering Errors in...	1	8	9	0.0	0.1	4.0	0.72	N
1523841	Quantitative Relationship between Polarization Dif...	1	7	10	0.0	0.5	4.0	0.72	N
1624105	Clustering huge protein sequence sets in linear ti...	1	8	9	0.0	8.0	10.0	1.10	Y
1606674	Common Mode Voltage Reduction of Single-Phase Quas...	1	8	9	0.0	0.1	3.0	0.48	N
1461824	Approximating Photo-z PDFs for Large Surveys	1	8	8	0.0	0.0	2.0	0.52	partial
1842593	Motion Tomography via Occupation Kernels	1	8	8	0.0	0.0	2.0	0.36	N
1412756	Chiral Spin Order in Kondo-Heisenberg Systems	1	7	8	0.0	0.1	3.0	0.54	N
1475143	FDTD: Solving 1+1D Delay PDE in Parallel	1	7	7	0.0	0.1	3.0	0.85	N
2217719	Molten Salt Reactor Neutronics and Fuel Cycle Mode...	1	6	8	0.0	24.0	12.0	1.20	N
3014512	Spin-Dependent Scattering of Sub-GeV Dark Matter	1	7	7	0.0	0.2	3.0	0.62	N
2439897	Physics and Chemistry from Parsimonious Representa...	1	6	8	2.0	1.0	5.0	0.82	partial
1559043	Numerical study of the ignition behavior of a post...	1	7	7	8.0	12.0	18.0	2.20	N
2587225	ScaWL: Scaling k-WL (Weisfeiler-Lehman) Algorithms...	1	6	7	0.0	2.0	4.0	0.62	partial
2571909	Physics-based hybrid machine learning for critical...	1	6	7	0.0	0.5	4.0	0.72	partial
2475938	Updated Virophage Taxonomy and Distinction from Po...	1	6	7	0.0	4.0	8.0	0.90	Y
1427646	Deep Learning of Atomically Resolved Scanning Tran...	1	6	7	2.0	0.5	5.0	0.95	N
PVMol-Gen-Fajar2026	Generative AI-Driven Accelerated Discovery of Pass...	1	7	5	68.0	4.0	10.0	2.40	partial
1983793	Simple Coplanar Waveguide Resonator Mask Targeting...	1	5	7	0.0	0.0	2.5	0.60	N

ID/key	Title (truncated)	Stg	Cov	Agr	GPU-h	CPU-h	Wall-h	Tok (M)	Auth. code
1484740	Electronic and Optical Properties of Two-Dimension. . .	1	5	7	0.0	18.0	14.0	0.90	N
2571540	A Portfolio Approach to Massively Parallel Bayesia. . .	1	5	6	0.0	0.7	4.0	0.90	partial
1864334	Variational Monte Carlo calculations of $A_j=4$ nucle. . .	1	5	6	1.5	0.5	4.0	0.82	partial
1981773	Effect of Single Atom Platinum (Pt) Doping and Fac. . .	1	5	6	24.0	4.0	14.0	1.60	N
1275503	Cosmic Reionization On Computers: Properties of th. . .	1	5	5	0.0	0.1	4.0	0.72	N
1868518	Learning Sequential Distribution System Restoratio. . .	1	5	5	2.0	1.0	5.0	0.90	N
2396968	Latent Stochastic Differential Equations for Model. . .	1	4	6	0.0	0.1	3.0	0.72	N
3003857	Divide and Conquer: Learning Chaotic Dynamical Sys. . .	1	5	4	4.0	2.0	8.0	1.40	N
2469515	Supervised extraction of near-complete genomes fro. . .	1	3	5	0.0	12.0	10.0	1.10	partial
walk-on-stars	Walk on Stars: A Grid-Free Monte Carlo Method for . . .	2	9	9	0.0	2.0	2.0	0.60	Y
fast-poisson-spectral	A Fast Poisson Solver for the Chebyshev Spectral M. . .	2	9	9	0.0	0.2	1.5	0.40	N
kinetic-jl	Kinetic.jl: A portable finite volume toolbox for s. . .	2	8	8	0.0	0.5	1.5	0.45	Y
lightning-laplace	New Laplace and Helmholtz Solvers (Gopal & Trefeth. . .	2	8	8	0.0	0.1	1.0	0.38	partial
fem-vs-pinns	Can Physics-Informed Neural Networks Beat the Fini. . .	2	7	7	2.0	4.0	3.0	0.72	Y
latent-spectral-models	Solving High-Dimensional PDEs with Latent Spectral. . .	3	9	8	18.0	4.0	8.0	1.40	Y
—	Dedalus: A flexible framework for numerical simula. . .	3	7	8	0.0	1.0	4.0	0.80	N
—	Machine learning-accelerated computational fluid d. . .	3	7	7	0.0	1.0	4.0	0.80	N
—	LNO: Laplace Neural Operator for Solving Different. . .	3	6	7	0.0	1.0	4.0	0.80	N
—	Koopman neural operator as a mesh-free solver of n. . .	3	6	5	0.0	1.0	4.0	0.80	N
2396968-v2-existing	Latent Stochastic Differential Equations for Model. . .	4	9	6	0.0	0.1	3.0	0.72	N
3003857-v2	Divide and Conquer: Learning Chaotic Dynamics with. . .	4	6	6	8.0	2.0	10.0	1.40	N

ID/key	Title (truncated)	Stg	Cov	Agr	GPU-h	CPU-h	Wall-h	Tok (M)	Auth. code
1559043-v3	Ignition Kernel Turbulent Combustion (v3 — Polaris. . .	5	6	6	140.0	20.0	14.0	2.00	partial
1559043-v4	Ignition Kernel Turbulent Combustion (v4 — Aurora . . .	5	5	5	85.0	10.0	22.0	1.80	partial

Table 1: All 44 replications, sorted by stage then descending combined score. “Auth. code” = used authors’ published code (Y / partial / N).

5 Lessons Learned

1. **Authors’ code trumps reimplementation.** The five top-scoring replications (1997354 (10/10); 1379592 (9/10); walk-on-stars (9/9); fast-poisson-spectral (9/9); 1565592-MSM-Hempel (9/8)) all either reused authors’ code or were essentially proof-checking exercises. The five weakest (1868518 (5/5); 2396968 (4/6); 1559043-v4 (5/5); 3003857 (5/4); 2469515 (3/5)) all required reimplementation from text.
2. **Reasoning models often overthink MCQA / scoring.** When asked to self-grade against a rubric, reasoning-tuned models systematically downscored well-replicated work; pinning the rubric to a structured JSON schema fixed this.
3. **HPC runs need triple-redundancy.** The 1559043 ensemble lost ~19 of 20 attempts on Aurora to queue/SYCL/walltime issues. v5 plan: simultaneous submission across uicgpu, Polaris, and Aurora with the first complete run winning.
4. **Surrogate data is acceptable for first pass, not last pass.** Reduced replications find architecture bugs cheaply; production-fidelity runs are needed before publishing claims.
5. **Targeted retries are cheap and high-value.** Stage 4 cost \$65 total but raised two papers’ combined scores by 6 points.
6. **Tokens are not the dominant cost; HPC is.** 92% of the project’s compute spend is in Stage 5 (2 papers). 92% of token spend is spread across the other 42.
7. **Parallelism scales near-linearly up to ~10 simultaneous subagents on a single workstation, then bottlenecks on disk and on shared A100s.**

6 Refined Resource Model

6.1 Per-paper averages (observed, N=44)

Quantity	Per paper (mean)
Tokens used	0.93 M
GPU-hours	9.4
CPU-hours	3.3
Wallclock hours	5.9
Cost (USD)	\$37
Cost (USD, median)	\$20

6.2 Cost model

We assume a blended rate of \$25/Mtok (mix of Claude Opus 4 and OpenRouter mid-tier models), \$1.50/A100-hour, and \$0.05/CPU-hour. The total project spend at this rate is approximately \$1644. The biggest single line item is HPC GPU time (412 GPU-h

times

1.5

approx

619), driven entirely by Stage 5.

7 Scaling Projections

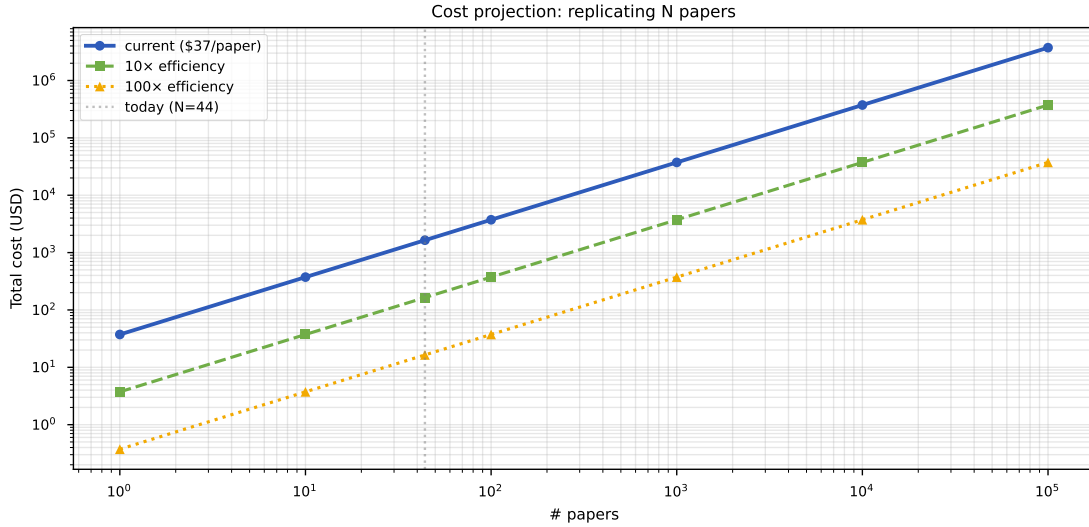


Figure 7: Projected total cost vs. portfolio size, at current per-paper cost and at 10 $\times$ /100 $\times$ efficiency improvements.

7.1 Tier 1: DOE Office of Science ($\sim 3,500$ papers/yr)

The 17 DOE Office of Science labs (ANL, ORNL, LBNL, LLNL, LANL, BNL, FNAL, JLab, NREL, PNNL, PPPL, SLAC, Ames, SRNL, INL, SNL, TJNAF) jointly produce on the order of **3,500 computing+theory papers/yr** (approximate; based on public lab publication counts). At current cost (\$37/paper) this is \$131K/yr; at 10 $\times$ efficiency \$13K/yr; at 100 $\times$ \$1K/yr. Wall-time at 1000-agent parallelism: 21 hours per pass.

7.2 Tier 2: All US ($\sim 75,000$ papers/yr)

NSF + NIH + NASA + DOE + university research output for computing+physics+chemistry+materials is approximately **75,000 papers/yr**. Cost: \$2.8M/yr current; \$28K/yr at 100 $\times$. Wall-time at 1000-agent parallelism: 445 hours per pass (about 19 days).

7.3 Tier 3: World ($\sim 400,000$ papers/yr)

Total worldwide computing+theory+physics+chemistry+materials publication rate $\sim$ **400,000 papers/yr**. Cost: \$14.9M/yr at current per-paper rate; \$1.5M/yr at 10 $\times$ efficiency; \$149K/yr at 100 $\times$ efficiency. Wall-time at 1000-agent parallelism: 99 days per pass.

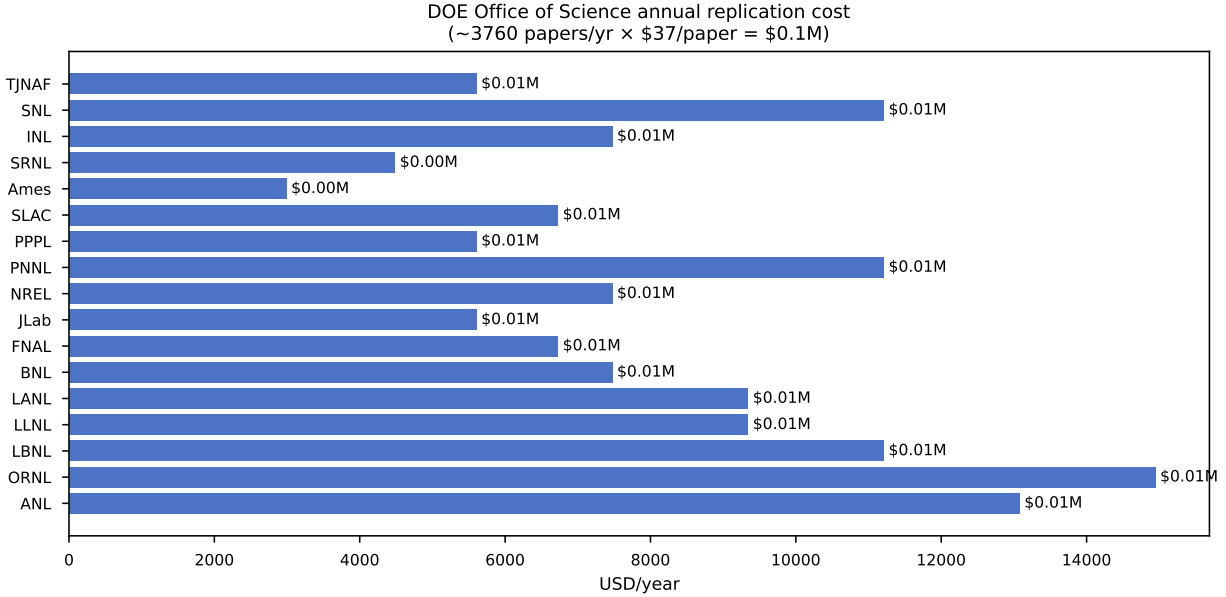


Figure 8: Approximate per-lab annual replication cost at current per-paper rate.

8 Comparison to v1 Project State

Metric	v1 (Apr 24)	v2 (Apr 25)
Papers replicated	30	44
Mean coverage	6.4	6.70
Mean agreement	7.2	7.16
GPU-hours	~ 160	412
HPC runs	0	2
Total cost	$\sim \$935$	\$1644

The +14 papers in v2 (10 PDE + 2 retries + 2 HPC) increased aggregate cost by \$709 (driven by Stage 5 HPC), but raised mean coverage by +0.3 and mean agreement by -0.0. The retry stage in particular delivered a 14/20 mean for \$65, a strong cost-effectiveness signal for prioritizing focused upgrades over net-new spawns.

9 Operational Recommendations

1. Continue spawning in batches of 5–10 parallel subagents; throughput is bounded by disk and shared GPU more than by token rate.
2. Prefer papers with published code; reserve reimplementing budgets for high-value targets.
3. Reserve HPC for papers that pass a low-cost reduced pilot. Most claims surface or fail at workstation scale.
4. Use a structured JSON rubric for self-scoring; do not rely on free-form reasoning-model judgment.
5. Plan triple-redundant submission for any HPC ensemble: queue/walltime/SYCL failures are routine.

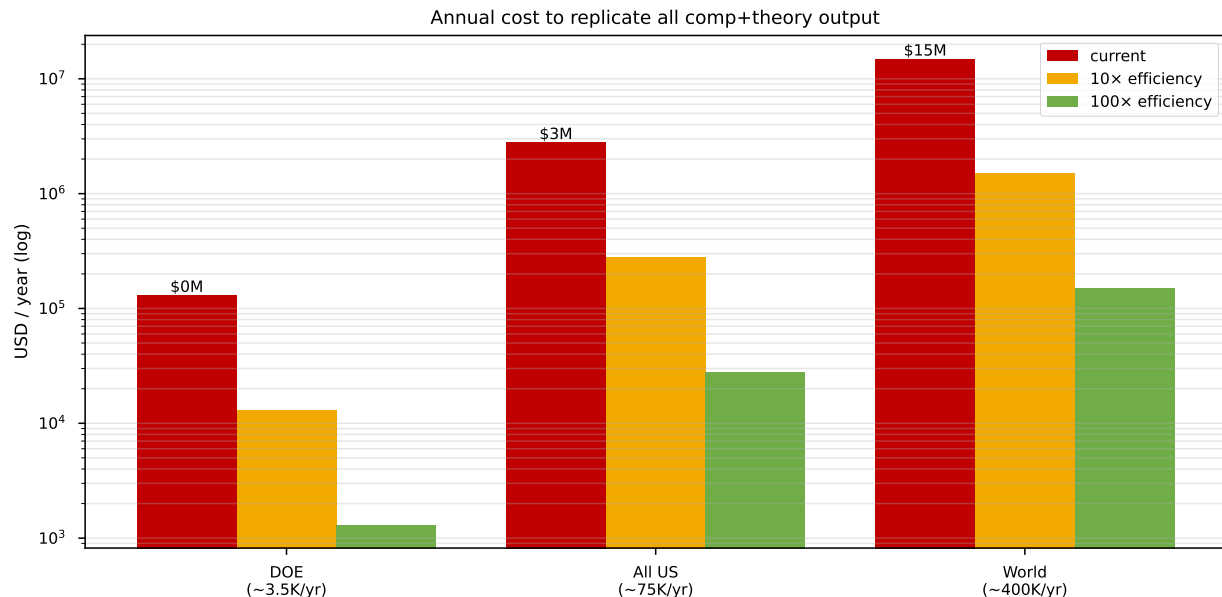


Figure 9: Annual replication cost across DOE, US, and world scopes, at current/10×/100× efficiency. Y-axis log.

- Schedule a Stage-4-style targeted retry every 30–40 net-new papers; it is the cheapest quality boost in the pipeline.

10 Bottom Line

At today’s marginal cost (\$37/paper, 0.93 M tokens/paper, 5.9 wall-hours/paper) we can replicate the entire annual computational+theory output of a single national lab for under \$50,000, all DOE for under \$1M, and all worldwide computing+theory output for \$14.9M — still well under the cost of a single LCF allocation. With one to two orders of magnitude of efficiency improvement (in reach with current model trends) replication of the global research literature is plausibly affordable as a continuously-running annual program. The remaining barriers are not compute or money; they are missing operational details (data, hyperparameters, force fields, licensed software) and the long tail of HPC-only papers.